Unit 5

Emerging Trends in Engineering Management

Participative Management

- It is also known as participative leadership or democratic leadership, is a management style that involves employees in decision-making and problem-solving.
- It's based on the idea that employees are close to the issues and can be just as legitimate as managers in proposing solutions.

Characteristics of Participative Management

- **Collaboration :** Employees are encouraged to work together transparently, sharing knowledge and skills.
- **Active participation :** Employees are encouraged to take an active part in decision-making.
- **Open exchange :** Managers act as facilitators or coaches, encouraging open exchange and listening to suggestions.
- **Consensus /Agreement :** Managers seek consensus before making important decisions.
- **Collective intelligence :** The aim is to harness collective intelligence and create an inclusive and motivating working environment.

Benefits of Participative Management

- Increased efficiency due to innovation and flexible feedback systems.
- Enhanced quality of products and lower manufacturing prices.
- Increased rate of achieving targets due to more frequent communication.
- Developed sense of ownership in employees.
- Decreased disputes between employees.

Conflict

- Conflict is a disagreement or clash between opposing forces, ideas, or interests.
- It can also refer to a mental struggle between incompatible wishes or drives.

Some examples of conflict:

- Two people have different ideas about how to use an empty lot
- Two people have opposing wants, such as a boss and an employee
- Two appointments are scheduled for the same time

- Conflict can be a natural part of the workplace, where different perspectives on business issues can lead to disagreements
- Conflict can be either good or bad. Controlled conflict is good for any organization’s growth.

Levels of conflict

A. **intrapersonal** - Conflict occurring within the self
B. **interpersonal** - This conflict May arise because of differences in goal and roll of an individual. This conflict occurs between two or more persons basically because of communication gap and perceptual differences.
C. **Intergroup** - it is occurred between groups because of difference between them. in sharing resources and implementation of conflicting rewards and punishment system may be taken as an example
D. **Inter-organizational**: It occurs between organizations because of various issues related to organizations functioning such as competition

Sources of Conflict

1. Personal differences
2. Goal and role incompatibility
3. Organizational climate and change
4. Gender and other social differences
5. Availability and accesses to resources and
6. Communication gap

Conflict Resolution

- It is a process to find a peaceful solution to a disagreement between two or more parties.
- It involves understanding different perspectives and working together to find a compromise that satisfies everyone.

How to Resolve Conflict

- **Avoidance** - This method suggests avoiding conflict by suppressing
- **Diffusion** - Conflict is tried to deactivate through diffusion.
- **Containment** - Conflict is allowed to bring into notice in a controlled manner and for resolution discussions and bargaining take place.
- **Confrontation** - Conflict related issues are brought out and tried to resolve the issues adopting necessary measures by addressing the needs of the parties in conflict. A face to face meeting conducted to find out the mutually agreed solutions.
- **Litigation / Legal Action** : pre defined law / rules and regulations are adopted to settle the disputes.

Change Management

- It's the process of implementing and describing changes within an organization.
- It involves:
  - Requesting changes
  - Determining if changes are attainable
  - Planning and implementing changes
  - Evaluating changes
  - Supporting people through the change process
  - Establishing goals and key performance indicators (KPIs)

Examples of change management

- Implementing new technology, Overhauling digital systems, Company acquisitions and mergers, and Changes in management personnel or style.
- When managing change, it's important to consider both hard and soft factors.
- Hard factors include:
  - The time required to complete the change
  - The number of people needed to execute the change
  - The financial results expected from the change

- Soft factors include:
  - Interpersonal skills: Compassionate listening, empathy, receptivity, trust, and self-awareness
  - Collaborative approach: A longer-term perspective that involves those affected by change
  - Mindset, culture, and behavior: Changes to the way people think and act, such as new corporate values, increased customer focus, or changes to diversity and inclusion

Quality Management

- Process of overseeing activities and tasks to maintain a desired level of excellence.
- It involves:
  - **Quality policy**: Determining a quality policy
  - **Quality planning and assurance**: Creating and implementing quality planning and assurance
  - **Quality control and improvement**: Managing quality control and improvement

Importance of QM

- Quality management is important for shaping company performance and customer satisfaction.
- Some types of Quality Management include:
  - **Total Quality Management (TQM):** Prioritizes meeting customer needs and exceeding expectations
  - **Six Sigma**: A methodology that helps businesses improve processes, products, or services by eliminating defects
  - **Quality Management System (QMS):** A system that can be broken down into four components: quality planning, quality control, quality assurance, and quality improvement

- The ISO 9000 family is a standardized set of rules for quality management systems.
- ISO 9001:2015 is a widely recognized quality management standard that outlines how an organization can achieve consistent performance and service.

Principles of Quality Management

- **Process approach**: Educating everyone involved in the production cycle about how inputs and outputs interrelate
- **Customer focus**: Ensuring everyone across the organization has a strong grasp on customers, suppliers, and competitors
- Leadership
- Engagement of People
- Continuous Improvement
- Evidence-based Decision Making
- Relationship Management

Benefits of Quality Management

- It helps an organization achieve greater consistency in tasks and activities that are involved in the production of products and services.
- It increases efficiency in processes, reduces wastage, and improves the use of time and other resources.
- It helps improve customer satisfaction.
- It enables businesses to market their business effectively and exploit new markets.
- It makes it easier for businesses to integrate new employees, and thus helps businesses manage growth more seamlessly.
- It enables a business to continuously improve its products, processes, and systems.

Innovation Management and Disruption

- Innovation management is the process of leading change and creating profits for a company, and disruptive innovation is a type of innovation that can radically change market behavior.
- Disruptive innovation can be a key component of innovation management, and companies need to have an innovation strategy in place to stay competitive.

Key concepts related to innovation management and disruptive innovation:

- **Disruptive innovation**: A new product or service that can create a new market segment or disrupt an existing one. Disruptive innovation can break existing operating models and create new ones.
- **Types of innovation**: Innovation can be incremental, breakthrough, or disruptive. Incremental innovation involves improving current products, services, processes, or methods. Breakthrough innovation is a technological advancement that improves a product or service within an existing category.
- **Disruptive innovation theory :**Popularized by Harvard Business School professor Clayton Christensen, this theory explains how new products, services, or technologies can start small and eventually surpass established offerings.

- **Examples of disruptive innovation**: Annapurna Conservation Area Project (ACAP) disrupted the traditional hotel industry by providing a platform for people to rent out their homes. The concept of Paying guest was developed.
- **Innovation management strategies**: Companies can use design thinking to structure and manage ideation, development, and innovation. They can also collaborate with external partners to solve innovation problems and reach goals more quickly.

Recent engineering management concepts for managing ICT based projects and organizations

Some engineering management concepts for managing ICT-based projects and organizations:

- **Project planning and strategy** : Ensure that project goals align with the organization's strategic objectives.
- **Resource allocation**: Allocate resources efficiently, including manpower, technology, and capital.
- **Team leadership and coordination** : Ensure that the team is led and coordinated effectively.

- **ICT strategy** : Have a well-laid out plan to coordinate activities to achieve the project's goals and mission.
- **Define, balance, and integrate relationships** : Define, balance, and integrate the relationships among the three vital elements of projects: People, Process, and technology.

- ICT project management is the process of planning, organizing, and developing information technology projects. IT project managers can use various software to plan and move through the various phases of the IT project management life cycle.

Thank You